

*Sleep Stage Patterns as Non-Invasive Biomarkers for Early Alzheimer's Disease
Risk Screening: A Machine Learning Approach*

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from BA889 for the Summer Semester of 2026*

August 3, 2026

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1. Introduction

Alzheimer's disease (AD) is one of the main neurodegenerative disorders and is responsible for 60 to 70% of the about 55 million cases of dementia in the world in 2023 [1]. Due to the ageing of the population, the number of patients is expected to go above 150 million by 2050, which will increase healthcare costs and place a greater strain on society [1]. Since pathological changes in the brain start decades before clinical symptoms appear, it is essential to carry out early detection—especially at the Mild Cognitive Impairment (MCI) stage—in order to make the most of treatment [2]. Nevertheless, the present standard diagnostic methods, including PET scans, MRI, and analysis of cerebrospinal fluid (CSF), are expensive, invasive, and not readily available, often resulting in diagnosis being made only after considerable cognitive decline has taken place.

In order to overcome these limitations, machine learning (ML)-based early screening, which makes use of data collected in a non-invasive manner, has become an increasingly viable alternative. The present study is concerned with the development of an ML classification model for the early detection of Alzheimer's disease by combining two publicly available datasets: cognitive assessment scores and clinical data from the Alzheimer's Disease Neuroimaging Initiative (ADNI), along with PSG-based sleep stage data from the Sleep Heart Health Study (SHHS). By applying algorithms including Logistic Regression, Random Forest, and XGBoost, we are able to differentiate between the Cognitively Normal (CN), MCI, and AD groups and to determine the most effective screening features. Finally, this study offers a cost-effective, non-invasive, and practical model that can assist healthcare companies, insurance providers, and medical institutions in identifying high-risk groups in advance.

2. Literature Review

2.1. The Biological Link Between Sleep and Alzheimer's Disease

Sleep is of great importance in the process of memory consolidation and in the removal of metabolic waste from the brain, a function mainly carried out by the glymphatic system [3],[4]. This network of perivascular pathways enables the exchange between cerebrospinal fluid (CSF) and interstitial fluid so as to eliminate neurotoxic substances, including amyloid-beta ($A\beta$) and tau proteins. The mechanism for this clearance is most active during slow-wave sleep (SWS), thanks to the dynamics of norepinephrine and AQP4-dependent fluid exchange [3]. As a result, sleep deprivation or sleep fragmentation leads to increased concentrations of these toxic proteins in the cerebrospinal fluid of humans and speeds up the development of Alzheimer's disease [4].

Beyond SWS, REM sleep is also important for memory processing and for Alzheimer's risk [5]. A systematic review published in 2024 of 17 longitudinal studies found that reduced REM sleep, lower sleep efficiency, sleep fragmentation, and sleep apnea were all consistent predictors of cognitive decline [6]. In particular, a smaller proportion of REM sleep hinders the consolidation of hippocampal-dependent memory and is an independent risk factor for developing dementia in the future [7].

2.2. Cognitive Assessment Tools and Early Screening for Alzheimer's Disease

In clinical practice, the Mini-Mental State Examination (MMSE) and the Alzheimer's Disease Assessment Scale-Cognitive Subscale (ADAS-Cog) are the two cognitive assessment tools most commonly used. The MMSE (with a maximum score of 30) is used as a quick screening method for cognitive decline, while the ADAS-Cog provides a more in-depth assessment of various cognitive areas

such as language, memory, and orientation. Both of these tools are able to quantitatively distinguish between the Cognitively Normal (CN), Mild Cognitive Impairment (MCI) and Alzheimer’s Disease (AD) groups, which is why they are important features in machine learning models.

A Random Forest model was recently suggested for the early detection of Alzheimer’s disease by an explainable machine learning method that makes use of multimodal clinical data [5]. The model, which included clinical variables such as MMSE scores, Activities of Daily Living (ADL) performance, and memory complaints, reached an AUC of 0.98 and had an accuracy rate of 95%. Moreover, the SHAP analysis found that MMSE and ADL were the most important features for classifying Alzheimer’s disease, thus verifying their clinical relevance as predictive factors.

2.3. Prior Research on Machine Learning-Based Alzheimer’s Classification

There has been a rapid increase in recent times in research aimed at the early detection of Alzheimer’s disease by means of machine learning, with a large number of studies based on the ADNI dataset having reported high performance when carrying out the three-class classification task for the Cognitively Normal (CN), Mild Cognitive Impairment (MCI), and Alzheimer’s Disease (AD) groups.

Studies that have used ADNI data have shown excellent results in the classification of three groups—CN, MCI, and AD—; an XGBoost model based on eight clinical features, such as MMSE, CDR, and ADAS-Cog, obtained a macro AUC-ROC of 0.982, and CDR and MMSE were found to be the most important predictors through SHAP analysis [8]. In a similar way, a balanced Random Forest developed from longitudinal survey data reached an accuracy of 88.5%, with ADL, age, and baseline MMSE being the most influential features [9]—this supports the rationale for feature selection used in the present study.

3. Data & Methods

3.1. Datasets

This study utilized two public datasets with distinct characteristics. Since the participant groups in the two datasets did not overlap, direct merging was not feasible; therefore, this study adopted an approach of conducting independent analyses for each dataset and then comprehensively discussing the combined results.

3.1.1. ADNI (Alzheimer’s Disease Neuroimaging Initiative)

ADNI is a large-scale, multi-center longitudinal study that has been ongoing since 2004, comprising five phases: ADNI1, ADNIGO, ADNI2, ADNI3, and ADNI4. This study utilized four CSV files containing cognitive assessment scores and diagnostic labels: Diagnostic Summary, Mini-Mental State Examination (MMSE), Alzheimer’s Disease Assessment Scale-Cognitive Subscale (ADAS-Cog), and Subject Demographics. Data were obtained from the official ADNI website following approval of the Data Use Agreement (<https://adni.loni.usc.edu>).

3.1.2. SHHS (Sleep Heart Health Study) / NSRR

SHHS is a large-scale, community-based cohort study conducted between 1995 and 2010 to investigate the association between sleep apnea and cardiovascular disease; it includes polysomnography (PSG) data from 5,804 adults aged 40 to 89. Access was granted via the National Sleep Research Resource (NSRR; <https://sleepdata.org>).

3.2. Data Preprocessing

3.2.1. ADNI Preprocessing

Only baseline visits (VISCODE = 'bl' or 'sc') were extracted from the longitudinal ADNI data and merged across four files using PTID, yielding 1,914 individuals. Following removal of records with missing diagnostic labels, group-wise median imputation of ADAS scores (14.5% missing), and exclusion of remaining incomplete rows, a final dataset of 1,887 participants (736 CN, 857 MCI, 294 AD) was obtained.

3.2.2. SHHS Preprocessing

Baseline data for 5,804 individuals were obtained by extracting only the first visit (visitnumber == 1) from the SHHS harmonized dataset (totaling 10,115 rows). Subsequently, to ensure demographic comparability with the ADNI cohort, the age range was restricted to 40–89 years, and rows containing missing values were removed. Ultimately, a total of 5,576 individuals were included in the analysis.

3.3. Feature Selection

3.3.1. ADNI Features

Correlation analysis revealed a Pearson correlation coefficient of 0.97 between ADAS_TOTAL and ADAS_TOTAL13, indicating extremely high multicollinearity. In light of this, ADAS_TOTAL13 was excluded, and the following five variables were ultimately selected as input features for the model. Detailed in Appendix A (Table A1).

3.3.2. SHHS Features

The following eight sleep variables, reflecting sleep architecture derived from polysomnography (PSG), were selected. These variables are key indicators consistently reported in previous studies on sleep biomarkers associated with Alzheimer's disease risk. Detailed in Appendix A (Table A2).

3.4 Classification Models

A three-class ML model (CN/MCI/AD) was built using ADNI data, while SHHS data supported exploratory sleep analysis. The dataset was split 80:20 into training (1,509) and validation (378) sets using stratified sampling. To address class imbalance (CN: 736, MCI: 857, AD: 294), `class_weight='balanced'` was applied, and all features were normalized via `StandardScaler`.

Three algorithms were evaluated: (1) Logistic Regression as a linear baseline (`'max_iter=1000'`); (2) Random Forest, an ensemble model enabling direct feature importance extraction, tuned via 5-fold `GridSearchCV` (`'max_depth=10'`, `'min_samples_split=5'`, `'n_estimators=200'`); and (3) XGBoost, a gradient boosting model effective for imbalanced and non-linear data (`'learning_rate=0.05'`, `'max_depth=3'`, `'n_estimators=100'`). Performance was evaluated using Macro AUC-ROC (One-vs-Rest) and class-specific Precision, Recall, and F1-score.

4. Results

4.1. ADNI Exploratory Data Analysis

A total of 1,887 ADNI participants were included in the final analysis following preprocessing; the diagnostic groups consisted of 736 cognitively normal (CN) individuals (39.0%), 857 with mild cognitive impairment (MCI) (45.4%), and 294 with Alzheimer’s disease (AD) (15.6%).

4.1.1. Distribution of Cognitive Function Scores

Figure 1 illustrates the distributions of MMSE and ADAS-Cog scores across the three diagnostic groups. Mean MMSE scores were 29.1 for CN, 27.5 for MCI, and 23.1 for AD, showing a consistent pattern of decline from the CN group to the AD group. Conversely, mean ADAS_TOTAL scores—which reflect cognitive decline—were lowest in the CN group and highest in the AD group, confirming that the two metrics measure cognitive function in opposite directions. There was some overlap in distributions among the three groups, suggesting that complete classification is difficult using a single metric alone.

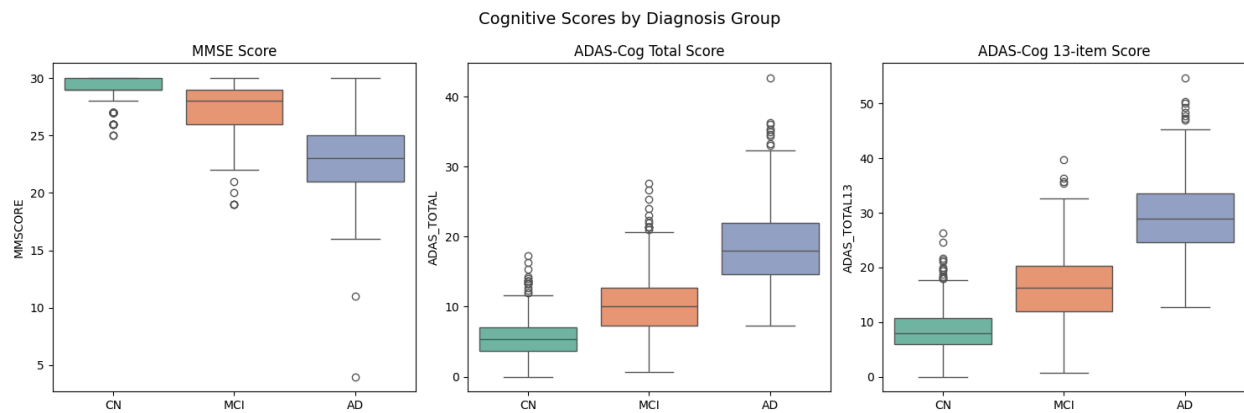


Figure 1: Cognitive Scores by Diagnosis

4.1.2. Demographic Characteristics

In the MCI group, there were significantly more males (506) than females (351), whereas in the CN group, there were more females (439) than males (301); the data thus reflect that gender is a factor associated with the onset of MCI. Median education levels were similar across the three groups at approximately 16 years, indicating that education level alone does not provide high discriminative power between the groups (Appendix A, Figure A1).

4.1.3. Age Distribution

The median age of the AD group (approximately 76 years) was higher than that of the CN (approximately 72 years) and MCI (approximately 74 years) groups, confirming a positive relationship between age and the risk of Alzheimer’s disease in the data. However, the age distributions of the three groups overlapped significantly, suggesting limitations in using age alone to distinguish between the diagnostic groups (Appendix A, Figure A2).

4.1.4. Feature Correlation Analysis

The correlation heatmap revealed strong negative correlations: the Pearson correlation coefficient was -0.70 between MMSCORE and ADAS_TOTAL, and -0.72 between MMSCORE and ADAS_TOTAL13. Notably, the correlation coefficient between ADAS_TOTAL and ADAS_TOTAL13

was 0.97; due to the confirmed presence of severe multicollinearity between these two variables, ADAS_TOTAL13 was excluded from the subsequent modeling stage (Appendix A, Figure A3).

4.2. SHHS Exploratory Data Analysis

Following preprocessing, a total of 5,576 SHHS participants (mean age: 63.01 years; mean BMI: 28.17) were included in the analysis.

4.2.1. Descriptive Statistics for Sleep Variables

Table 1 presents descriptive statistics (mean, standard deviation, minimum, and maximum) for key sleep variables across the entire cohort. On average, the mean proportion of REM sleep was 19.8%, slightly lower than the standard for healthy adults (20–25%), while the mean proportion of slow-wave sleep (Stage 3) was 18.2%, falling within the normal range (15–25%). The mean sleep efficiency was 82.9%, which is below the clinical norm of 85%, and the mean AHI was 14.6 events/hour, bordering the threshold for mild sleep apnea (5–15 events/hour). The mean sleep latency was 14.06 minutes, falling within the normal range (under 30 minutes) (Appendix A, Figure A4).

	nsrr_pctdursp_s1	nsrr_pctdursp_s2	nsrr_pctdursp_s3	nsrr_pctdursp_sr	nsrr_ttleffsp_f1	nsrr_ttlatp_f1	nsrr_ttlurws_f1	nsrr_ahi_hp3u	nsrr_phnumar_f1
mean	5.44	56.54	18.21	19.80	82.77	14.07	61.44	14.65	19.16
std	3.95	11.72	11.87	6.27	10.55	20.40	44.03	15.61	10.66
min	0.00	10.88	0.00	0.00	11.31	0.00	0.00	0.00	0.00
max	39.12	100.00	70.06	43.30	104.15	217.50	368.00	161.84	110.39

Table 1: Descriptive Statistics for Key Sleep Variables

4.2.2. Changes in Sleep Architecture by Age Group

Figure 2 illustrates changes in sleep architecture across age groups (40–59, 60–69, 70–79, and 80–89 years). A distinct pattern was observed wherein the proportions of Stage 3 (slow-wave sleep) and REM sleep decreased with advancing age; concurrently, a trend of reduced sleep efficiency and increased WASO (wake after sleep onset) was identified. These results indirectly support the study’s core hypothesis that age-related changes in sleep architecture coincide with the period of increased Alzheimer’s risk.

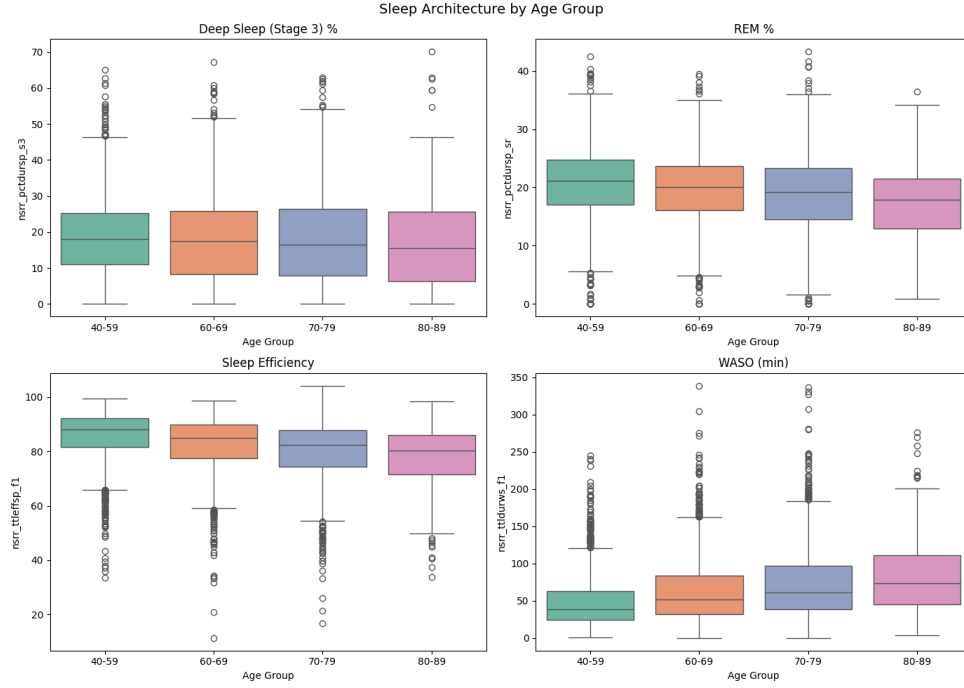


Figure 2: Sleep Architecture by Age Group

4.2.3. Correlation Analysis of Sleep Variables

Correlation analysis confirmed a strong positive relationship between AHI and the Arousal Index, and a strong negative relationship between WASO and sleep efficiency. A negative correlation between age and Stage 3 proportion numerically confirmed the age-group patterns above (Appendix A, Figure A5).

4.3 Classification Model Performance

4.3.1. Initial Model Performance Comparison

Based on initial training results, the macro AUC-ROC scores were 0.881 for Logistic Regression, 0.888 for Random Forest, and 0.885 for XGBoost. All three models achieved an AUC-ROC of 0.88 or higher, demonstrating performance levels considered clinically useful.

4.3.2. Hyperparameter Tuning Results

5-fold cross-validation using GridSearchCV identified the optimal parameters as 'max_depth=10', 'min_samples_split=5', and 'n_estimators=200' for Random Forest, and 'learning_rate=0.05', 'max_depth=3', and 'n_estimators=100' for XGBoost. Following tuning, performance improved to an AUC-ROC of 0.8925 for Random Forest and 0.8906 for XGBoost.

4.3.3. Final Model Performance Comparison

Table 2 presents the final performance comparison of the models. Random Forest (Tuned) achieved the highest performance with an AUC-ROC of 0.8925 and was selected as the final model.

Model	AUC-ROC	Accuracy
Logistic Regression	0.881	0.75
Random Forest	0.888	0.73
XGBoost	0.885	0.73

Table 2: Final Model Performance Comparison

The feature importance analysis revealed that MMSCORE and ADAS_TOTAL had the highest importance, indicating that cognitive assessment scores play the most critical role in classifying CN, MCI, and AD. AGE showed moderate importance, while PTEDUCAT and PTGENDER exhibited relatively low importance. Performance comparison of the final models and the feature importance of the tuned Random Forest model is detailed in Appendix A (Figure A6).

4.3.4. Detailed Performance by Class (Tuned Random Forest)

Figure 3 shows the ROC curves by class for the final model.

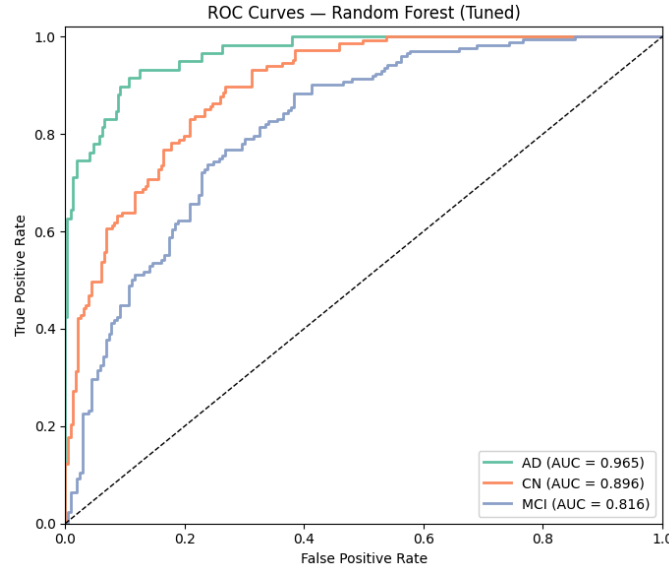


Figure 3: ROC Curves by Class

The recall for the AD group was 0.76, indicating that 76% of actual Alzheimer’s patients were accurately detected. The CN group achieved the highest recall among the three classes at 0.80. The MCI group had the lowest F1-score at 0.69, reflecting the fact that MCI is a heterogeneous group with characteristics intermediate between CN and AD. Detailed performance by class of tuned Random Forest is in Appendix A (Table A4).

5. Discussions & Limitations

5.1. Discussion

5.1.1. Clinical Significance of Model Performance

The final selected Random Forest (Tuned) model achieved a macro AUC-ROC of 0.8925. Given that an AUC-ROC of 0.8 or higher is generally considered clinically useful—with values closer to 0.9 deemed excellent—this model demonstrated performance suitable for practical application in the three-class classification task (CN/MCI/AD). While this performance is slightly lower than the AUC-ROC of 0.982 reported by Esan et al. [8] using XGBoost on the same ADNI dataset, the results are significant considering that the previous study utilized eight clinical features (including the Clinical Dementia Rating, or CDR), whereas the current study relied on a more limited set of only five non-invasive features.

5.1.2. Feature Importance and Clinical Interpretation

Feature importance analysis revealed that MMSCORE and ADAS_TOTAL held the highest importance. This indicates that cognitive assessment scores are the strongest predictors for Alzheimer's classification, a finding consistent with prior studies by Aghalari et al. [5] and Ren et al. [9], which identified MMSE and activities of daily living (ADL) as the most critical features. Age (AGE) showed moderate importance, suggesting it provides complementary predictive value when used alongside cognitive function scores.

5.1.3. Link Between Sleep Data and Cognitive Function

Analysis of the SHHS data revealed a consistent pattern in which the proportions of slow-wave sleep (Stage 3) and REM sleep decreased with advancing age. This aligns with the pattern observed in the ADNI dataset—where increasing age correlates with lower MMSE scores and a higher risk of Alzheimer's—and indirectly supports the argument from previous studies [5][6] that changes in sleep architecture may serve as potential biomarkers for cognitive decline. Given Pase et al.'s finding that each 1% decrease in REM is associated with a 9% increase in dementia risk [7], the cohort's mean REM of 19.8%—below the normal range of 20–25%—suggests a non-trivial proportion of high-risk individuals within the SHHS sample.

5.2 Limitations

First, the two datasets could not be directly merged due to non-overlapping participants, preventing sleep variables from being used as direct model features. Furthermore, the absence of cognitive diagnostic labels in SHHS precluded direct statistical verification of the sleep-cognition relationship. Future work utilizing cohorts such as the MrOS Sleep Study or Framingham Heart Study—where both PSG and cognitive assessments were collected—would address this limitation.

Second, limitations in MCI classification performance. In the final model, the F1-score for the MCI group was 0.69, which was lower than that of the CN (0.75) and AD (0.77) groups. This reflects the clinical characteristic that MCI lies on a cognitive continuum between CN and AD, making it difficult to clearly distinguish MCI from the other groups based solely on cognitive scores from a single time point. Incorporating longitudinal data on cognitive changes or additional biomarkers could improve MCI detection performance.

Third, the issue of the ADNI cohort's representativeness. The ADNI cohort is characterized by a high average level of education (16 years, equivalent to a university degree) and limited racial diversity. This may constrain the model's generalizability; performance could vary if the same model is applied to populations with lower education levels or diverse racial and cultural backgrounds.

Fourth, the measurement accuracy of consumer-grade wearable devices. While the ultimate goal of this study is screening using sleep data collected from consumer wearables such as smartwatches, the SHHS data used here is based on measurements from clinical polysomnography (PSG) equipment. Currently, consumer wearables exhibit lower accuracy in classifying sleep stages compared to clinical PSG, with particular limitations in distinguishing between Stage 3 and REM sleep. Therefore, further validation is required before the study's findings can be directly applied to data from wearable devices.

6. Conclusion

This study developed a machine learning-based classification model for the early screening of Alzheimer's disease and explored the potential of sleep architecture data as a non-invasive biomarker. To this end, clinical cognitive data from ADNI (1,887 subjects) and sleep polysomnography (PSG) data from the SHHS dataset via NSRR (5,576 subjects) were analyzed independently.

A three-class classification model (CN/MCI/AD) was constructed using five features—MMSE, ADAS-Cog, age, years of education, and gender—based on the ADNI data. A Random Forest model optimized via GridSearchCV achieved a macro AUC-ROC of 0.8925, demonstrating clinically significant performance. Feature importance analysis identified MMSE and ADAS-Cog as the strongest predictors, reaffirming the pivotal role of standardized cognitive assessment tools in Alzheimer's screening. The SHHS analysis revealed consistent patterns of declining slow-wave sleep (Stage 3) and REM proportions, as well as reduced sleep efficiency, with advancing age; these trends aligned with the age-related cognitive decline observed in the ADNI data, indirectly supporting the hypothesis that changes in sleep architecture could serve as a potential biomarker for cognitive decline.

The primary significance of this study lies in demonstrating the feasibility of a practical screening model capable of identifying high-risk Alzheimer's groups—without the need for costly or invasive diagnostic procedures—using only routinely collected cognitive assessment scores and sleep data. This provides a practical foundation for healthcare companies, insurance providers, and medical institutions to conduct low-cost, large-scale preliminary screening for Alzheimer's risk. Future research should establish an integrated cohort containing both sleep PSG data and Alzheimer's diagnosis labels for the same participants to develop a more robust screening model that directly incorporates sleep variables as features. Furthermore, applying and validating this model using real-world sleep data collected via consumer wearable devices would bring us closer to its practical implementation as an accessible early screening tool for Alzheimer's disease in everyday settings.

7. References

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Appendix A.

Table A1: Feature Selection Table (ADNI)

Variable Name	Description
MMSCORE	Mini-Mental State Examination score (0–30; higher scores indicate normal function)
ADAS_TOTAL	ADAS-Cog total score (0–70; higher scores indicate greater cognitive impairment)
AGE	Age at the time of assessment (years)
PTEDUCAT	Years of education (years)
PTGENDER	Gender (Male=1, Female=0)

Table A2: Feature Selection Table (SHHS)

Variable Name	Description
nsrr_pctdursp_s1	Stage 1 (Light Sleep) Percentage (%)
nsrr_pctdursp_s2	Stage 2 (Intermediate Sleep) Percentage (%)
nsrr_pctdursp_s3	Stage 3 (Slow-Wave Sleep, Deep Sleep) Percentage (%)
nsrr_pctdursp_sr	REM Sleep Percentage (%)
nsrr_ttleffsp_f1	Sleep Efficiency (%)
nsrr_ttlatasp_f1	Sleep Latency (min)
nsrr_ttdurws_f1	Wake After Sleep Onset (WASO, min)
nsrr_ahi_hp3u	Apnea-Hypopnea Index (AHI, events/hour)

Figure A1: Gender Distribution and Education Level by Diagnostic Group (ADNI)

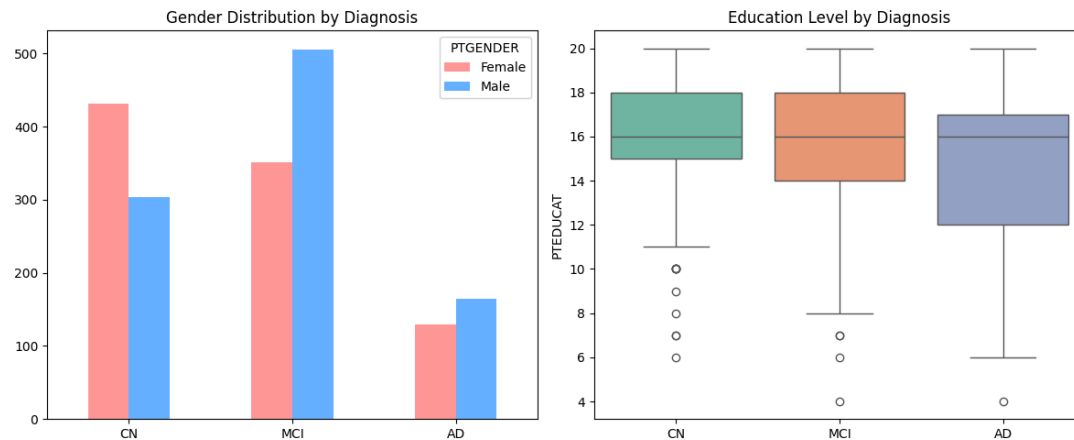


Figure A2: Age Distribution by Diagnostic Group (ADNI)

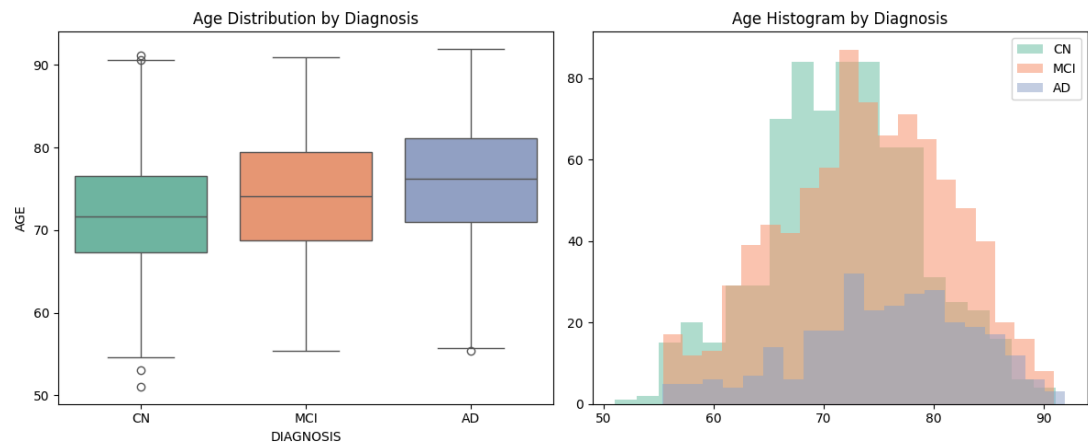


Figure A3: Feature Correlation Heatmap (ADNI)

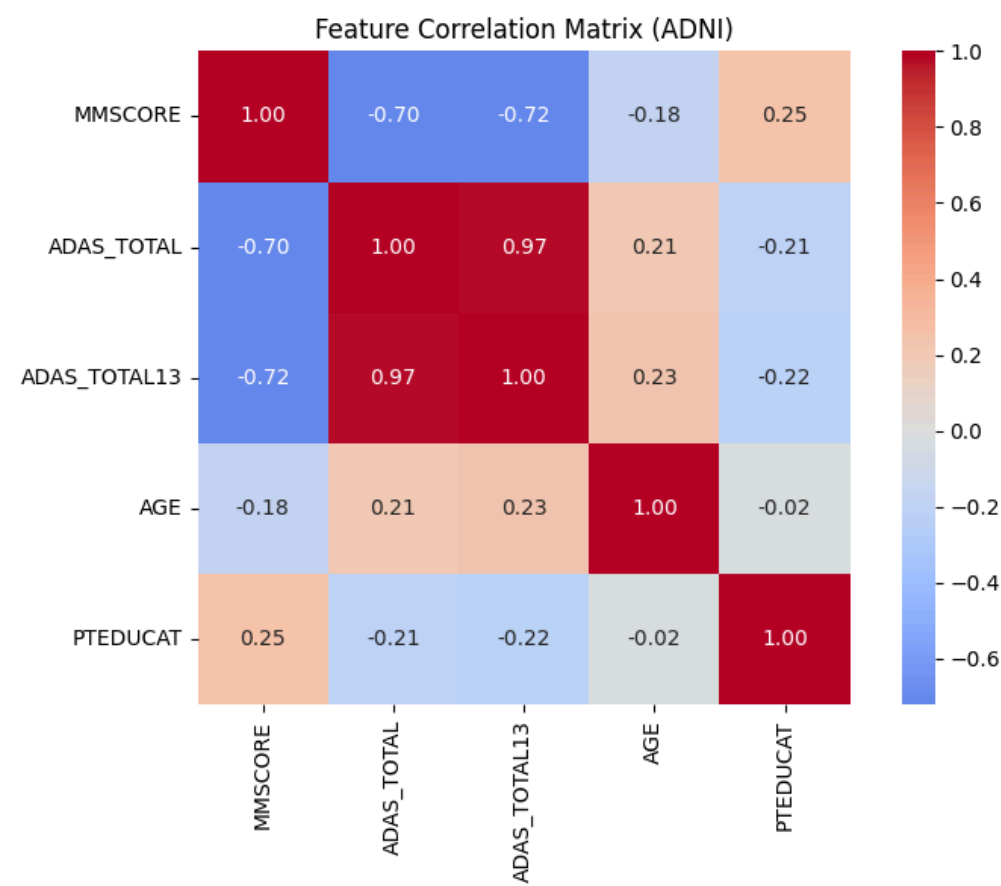


Figure A4: Distribution of Each Sleep Variable (SHHS)

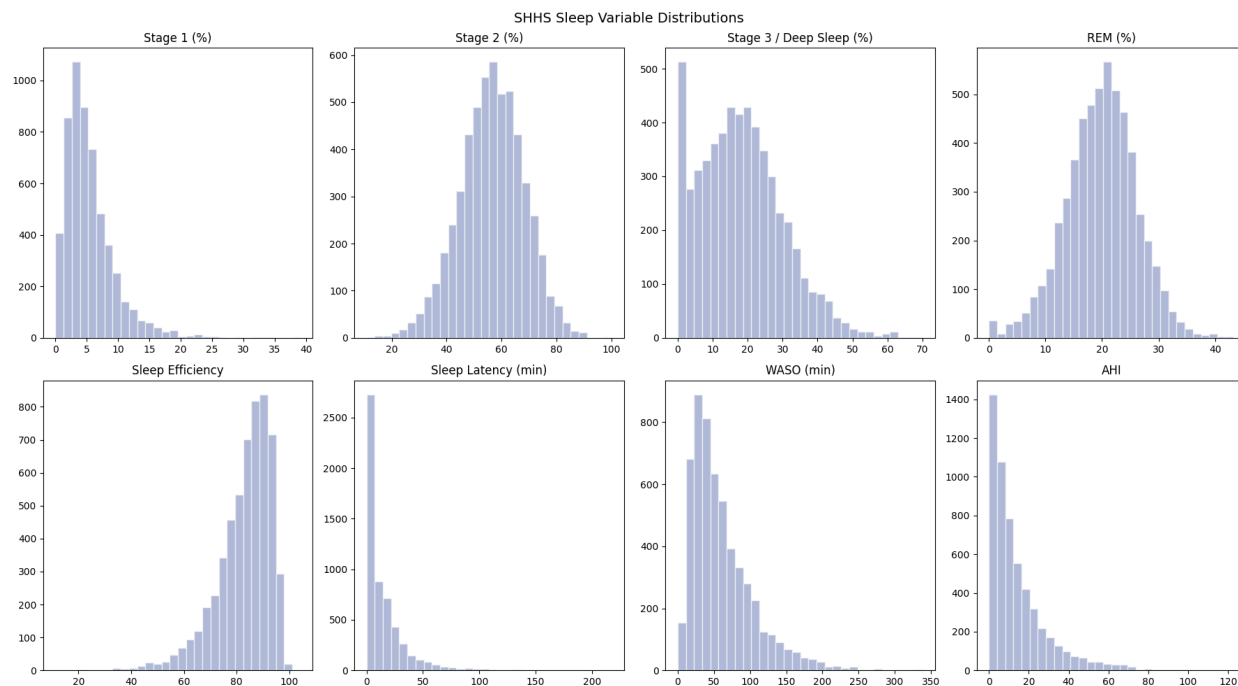


Figure A5: Sleep Feature Correlation Heatmap (SHHS)

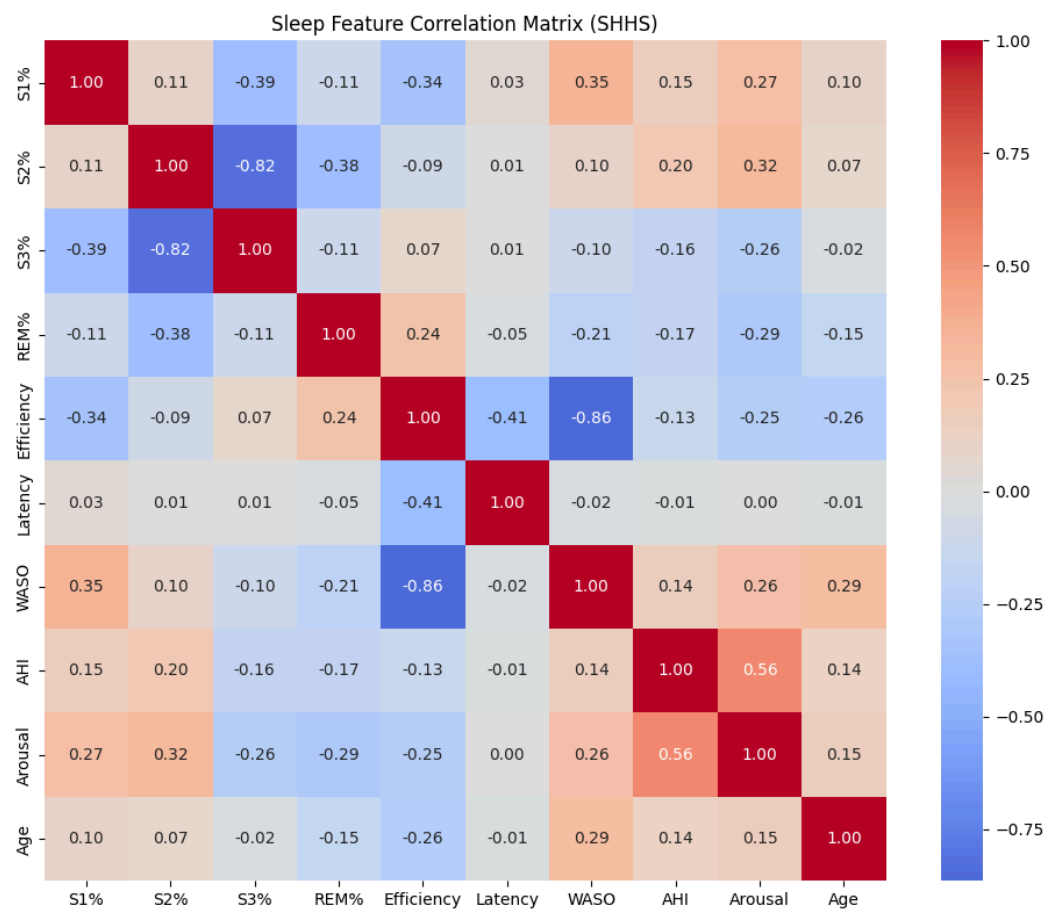


Table A3: Final Model Performance Comparison

Model	AUC-ROC
Logistic Regression	0.881
Random Forest	0.888
XGBoost	0.885
Random Forest (Tuned)	0.893 ← Best
XGBoost (Tuned)	0.891

Figure A6: Performance Comparison & Feature Importance of the Tuned Random Forest

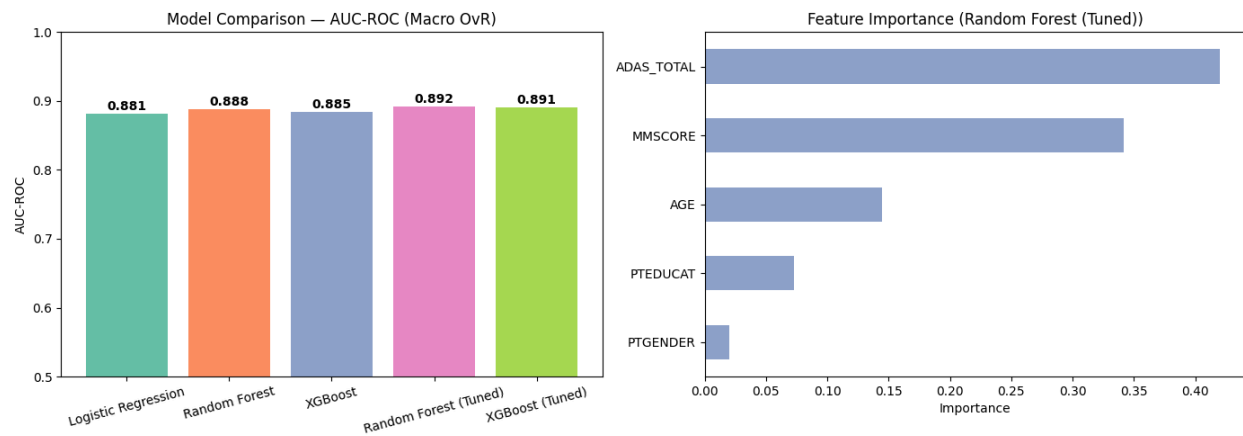


Table A4: Performance by Class of Tuned Random Forest

[Random Forest (Tuned)]		AUC-ROC: 0.8925			
	precision	recall	f1-score	support	
AD	0.78	0.76	0.77	59	
CN	0.72	0.80	0.75	147	
MCI	0.72	0.66	0.69	172	
accuracy			0.73	378	
macro avg	0.74	0.74	0.74	378	
weighted avg	0.73	0.73	0.73	378	